

UNIVERSITY OF ESSEX

Postgraduate Examinations 2024

PROFESSIONAL PRACTICE AND RESEARCH METHODOLOGY

Time allowed: **TWO** hours

This exam is Open Book (Restricted), students may bring/use the following in the exam:
Study Notes (max 6 double-sided A4 pages)

Candidates are permitted to use:
Casio FX-83GT or Casio FX-85GT Series **ONLY** (CW, X, Plus)

The following items are available:
Scrap paper (on the invigilator's desk)

The paper consists of **NINETEEN** questions. Candidates must answer **ALL** questions. There are 17 multiple choice questions in **SECTION A** and 2 written answer questions in **SECTION B**. Answers to Section B should be typed directly into Moodle.

Questions in Section A are of equal weight, questions in Section B are **NOT** of equal weight.

Section A is worth 51% of the paper, Section B is worth 49%.

Please do not leave your seat unless you are given permission by an invigilator.

Do not communicate in any way with any other candidate in the examination room.

If you have a query with the content of this exam paper, or if you have a technical problem with MOODLE, or any other query, please ask an invigilator.

NOTE that you will be marked on your FIRST ATTEMPT only, unless there are technical reasons why you need to start another attempt.

At the end of the examination, remain seated until you have been told you may leave.

Your responses must be your own work. Procedures are in place to detect plagiarism.

SECTION A

SECTION A IS NOT AVAILABLE

SECTION B

All ANSWERS to SECTION B must be written in MOODLE

Question 18

- a) If you were recruiting for the following role for a Software Engineer, list four things that you would likely be looking for in the successful candidate. [3%]

Our company is a global leader in industrial software. Our cutting-edge solutions are used by thousands of enterprises to deliver the essentials of life – such as energy, infrastructure, chemicals and minerals – safely, efficiently and more sustainably. If you're a curious and collaborative person who wants to make a big impact through technology, then we want to hear from you!

Position: Software Engineer

Previous experience: Experience in software development, ideally with a focus on C#, .NET Core, and Azure-based solutions.

Location: Cambridge, United Kingdom – Hybrid (60% Office, 40% Home)

Employment type: Full-time regular (flexible working options available)

The Role: As a Software Engineer, you'll partner with a team of dedicated software developers to design and develop an Azure solution. This solution will administer and manage the industrial design data that underpins our design products. You will collaborate with Product Owners, Stakeholders, and the Scrum Team in researching, designing, and implementing complex technical solutions, leveraging cloud and edge technology to provide a highly available global solution for managing industrial design data.

Key Responsibilities:

- *Work with Product Owners, Stakeholders, and Scrum Team to research, design, and implement complex technical solutions.*
- *Develop and maintain robust cloud solutions using ASP.Net Core and C#.*
- *Engage in Scrum meetings, including sprint planning and estimating; backlog refinement; daily Scrum meetings; sprint retrospectives and sprint reviews.*
- *Collaborate with a diverse range of people, including Product Owners, Testers & Product Support across sites.*
- *Coach and mentor less experienced colleagues and foster a positive, collaborative work environment.*

- b) When a job advertisement invites you to apply by submitting an application and CV, the application is generally in the form of a covering letter accompanying the CV. Briefly outline the structure and content that would be expected in a competent application covering letter. [7%]

Question 19

The formula for Present Value of money is:

$$\frac{X}{(1 + r)^t}$$

Where X is the amount to be adjusted, r is the interest rate as a fraction and t is the time period.

- a) Your cousin needs £1,000 now and promises to pay you back £1,150 in 2 years. Is that a good investment compared to an alternative option where you can earn 5% per annum? Please justify your answer using the Net Present Value (NPV) calculation. [7%]
- b) Will the investment decision change if you can get 12% elsewhere? Please justify your answer with an NPV calculation. [7%]

Question 20

Two project engineers in a company propose 2 projects. The company needs to make a decision on which project it should invest in. The company raises \$5 million cash at 9% interest rate for the initial investment (which needs to be returned at 9%) in one of the projects. The cash flow of the 2 proposed projects A & B can be seen in the table below:

Project A		Discounted cash Flow
Year	Cash Flow	
0	-\$5 million, initial investment	
1	\$3 million profit	
2	\$2 million profit	
3	\$4 million profit	
4	\$3 million profit	
5	\$0, project closeout	

Project B		Discounted cash Flow
Year	Cash Flow	
0	-\$5 million, initial investment	
1	\$0	
2	\$0	
3	\$0	
4	\$0	
5	\$16 million profit	

- a) Compute the Discounted Cash Flow for both projects and fill in the table above for all the years (0-5). [14%]
- b) Calculate the net present value (NPV) of both projects A & B? [8%]
- c) Which project, assuming both carry the same risk, should the financial analyst recommend to the manager? Please explain your reasons in a few sentences. [3%]

END OF SECTION B

END OF PAPER CE902-7-SP